

Detailed explanation of the operation method and industry application of the AILEIKE ALKC400D precision digital pressure gauge

The ALKC400D precision digital pressure gauge (with temperature compensation) boasts an accuracy level of 0.05. Its 5-digit display provides a highly visible and intuitive readout, complemented by a backlight that ensures normal operation in poorly lit environments. The sealed housing structure ensures excellent protection, and the device offers advantages such as stable numerical display, rapid response speed, and clear digital readout. This equipment serves not only as a standard instrument for metrological calibration but also as a high-precision pressure monitoring instrument for long-term use in industrial sites. It stands as a precision device in the field of pressure detection and calibration.

1. Technical indicators

Parameter item	Technical indicators	Parameter item	Technical indicators
pressure range	-100kPa to 0 to 100MPa (range selectable) with options for gauge pressure, absolute pressure, and micro pressure	Accuracy (or Precision)	Level 0.05
Temperature compensation range	Ensure instrument accuracy within the range of 0~50°C	resolution	Pressure: min 1 Pa (depending on the measurement range)
overload capacity	1.2 to 1.5 times the upper limit of the measurement range	Display	LCD backlit display; 5-digit display
working environment	Temperature: 0~50°C	relative humidity	≤80%RH
Overall dimensions/weight	Outer diameter of the instrument: Φ100mm / 600g	Power supply	External charger 220VAC

III. Operating Method

1. "PEAK": Peak value button. If in use, pressing it once will display the maximum value from the current measurement process and show "HI", while pressing it again will display the minimum value and show "LO". When displaying the maximum/minimum values, pressing the "ZERO" button can reset them to zero respectively.

2. "ZERO" zero calibration button, which can reset to zero with one click under no pressure condition.

3. "P/A": Back button. Pressing the "P/A" button will return to the main pressure unit.

4. "UNIT": Unit Switch Button. When in use, press the "UNIT" button to toggle between six units (kPa, MPa, psi, mmHg, bar, kgf/cm²) (the specific unit depends on the measurement range).

5. "Backlight": Press the backlight button to turn the backlight on or off.

IV. Application Scenarios

1. Metrology Laboratory/Third-Party Testing: Serving as a pressure standard gauge, it is used in conjunction with a pressure calibration bench to verify ordinary pressure gauges, vacuum gauges, and precision pressure gauges

2. Oil and gas, energy industry: on-site comparison and calibration of pressure instruments for oil and gas wellheads and gas pipelines

3. Chemical, pharmaceutical, and rubber and plastic manufacturing: precise pressure measurement of clean pipelines and hydraulic/pneumatic process circuits

4. Research institutes and university laboratories: teaching metrology training, standard equipment for pressure metrology experimental teaching

5. Special equipment, water supply, and fire protection fields: On-site verification of pressure instruments for municipal water supply and fire protection pipe networks

V. Precautions

1. The instrument should be placed in a dust-free and non-corrosive environment.
2. When the instrument is not in use, the power should be turned off. If it is not used for a long time, it should be charged once every three months.
3. The instrument has sufficient overload capacity, but do not overload it severely.
4. The instrument should be handled with extreme care, especially during on-site use, to avoid falling or impact.
5. The silicon diaphragm of the sensor is very thin, so it must not collide with solid objects.

VI. Conclusion

The AILEIKE ALKC400D precision digital pressure gauge boasts advantages such as a wide measurement range, portability, and durability. It balances metrological calibration performance with practicality for industrial sites, making it suitable for various pressure measurement scenarios including metrological verification, industrial inspection, scientific research, and quality control. The device is easy to operate and offers good stability, enhancing the efficiency of pressure detection work. This series of digital pressure gauges can be used for the verification, calibration, and testing of gauge pressure, absolute pressure, and vacuum (negative pressure), as well as for precise pressure measurement.